

Trigonometric functions and equations - Set 2 - Solutions

Specification MM4.3-MM4.6.

Answer key

Question	Answer	Topic
1	E	Trigonometric functions and equations
2	E	Trigonometric functions and equations
3	E	Trigonometric functions and equations
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Worked solutions

Question 1 X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III For every real k the equation has at least one solution.

Answer: E

I and II are read from the graph of $\sin$ over one period. III is false: $\sin x = 2$ has no solutions.

Full method - how and why

Read the claims off the graph of $y = \sin x$ over one period, then remember the range.

- 1 I: for $-1 < k < 1$ the horizontal line $y = k$ cuts one period of the sine curve exactly twice. **True.**
why: Sine rises from 0 to 1, falls to -1 and returns - a strictly-between value is hit once on the way down and once elsewhere in $[0, 2\pi)$.
- 2 II: $k = 1$ is reached only at the single peak $x = \frac{\pi}{2}$. **True.**
why: The maximum is attained once per period.
- 3 III: for $|k| > 1$ there is no solution at all - $\sin x = 2$ is impossible. **False.**
why: The range of sine is $[-1, 1]$.
- 4 So I and II.
why: Match to the list.

Where it goes wrong

- Accepting III by forgetting the range restriction.
- Rejecting II by counting $x = \frac{\pi}{2}$ and $x = \frac{5\pi}{2}$ - the second is outside $0 \leq x < 2\pi$.
- Rejecting I because $k = 0$ feels special: $\sin x = 0$ gives $x = 0$ and $x = \pi$ in the interval - still exactly two.

Takeaway Per period, $\sin x = k$ has two solutions for $|k| < 1$, one for $|k| = 1$ and none for $|k| > 1$. Sketch the line across the curve.

Question 2 TRG-02 · Trigonometric functions and equations · spec MM4.5

Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

Answer: E

$2(1 - \sin^2 x) + 3\sin x - 3 = 0 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0$. $\sin x = \frac{1}{2}$: 30° , 150° ; $\sin x = 1$: 90° . Three solutions.

Full method - how and why

A mixed sin/cos equation. Use $\cos^2 = 1 - \sin^2$ to make it a quadratic in $\sin x$, then count solutions carefully.

1 Replace $\cos^2 x$ with $1 - \sin^2 x$: $2(1 - \sin^2 x) + 3\sin x - 3 = 0$.

why: The identity $\sin^2 + \cos^2 = 1$ is the only way to get a single trig function - always convert the squared one.

2 Tidy: $-2\sin^2 x + 3\sin x - 1 = 0$, or $2\sin^2 x - 3\sin x + 1 = 0$.

why: Multiply through by -1 so the leading coefficient is positive.

3 Factorise: $(2\sin x - 1)(\sin x - 1) = 0$, giving $\sin x = \frac{1}{2}$ or $\sin x = 1$.

why: Treat $\sin x$ as a single unknown.

4 Count solutions in $0^\circ \leq x \leq 360^\circ$: $\sin x = \frac{1}{2}$ gives 30° and 150° ; $\sin x = 1$ gives only 90° .

why: A maximum value like $\sin x = 1$ is reached just once per period, not twice - this is what makes the total odd.

5 Total: 3.

why: Count, do not assume 'two roots means four solutions'.

Where it goes wrong

- Assuming each value of $\sin x$ gives two solutions and answering 4.
- Sign slips when substituting the identity, producing an unfactorisable quadratic.

Takeaway Convert to one trig function, factorise, then count solutions per value - endpoints ± 1 give only one each.

Question 3 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of $y = \cos x$ is/are true?

I It is symmetric about the y-axis.

II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.

III It has period π .

Answer: E

$\cos(-x) = \cos x$ so I is true. $\sin(x + \frac{\pi}{2}) = \cos x$, so II is true. The period of $\cos$ is 2π , so III is false.

Full method - how and why

Three properties of the cosine graph. Each is a standard fact; the false one is about the period.

1 I: $\cos(-x) = \cos x$, so the graph is unchanged by reflection in the y-axis. **True.**

why: Cosine is an even function - its graph is symmetric about the y-axis, unlike sine.

2 II: $\sin(x + \frac{\pi}{2}) = \cos x$, and replacing x by $x + \frac{\pi}{2}$ shifts the graph $\frac{\pi}{2}$ to the left. **True.**

why: Check a point: $\sin$ at $x = \frac{\pi}{2}$ equals 1, and $\cos$ reaches 1 at $x = 0$ - shifted left by $\frac{\pi}{2}$ ✓.

3 III: the period of $\cos$ is 2π , not π . **False.**

why: $\cos(x + \pi) = -\cos x$, so a shift of π flips the graph rather than repeating it.

4 I and II only.

why: Match to the option list.

Where it goes wrong

- Thinking the period is π because $\cos$ returns to zero every π - the **whole** shape must repeat.
- Getting the direction of the shift in II backwards.

Takeaway $\cos$ is even with period 2π ; $\cos x = \sin(x + \frac{\pi}{2})$ is a shift to the left.

Question 4 TRG-14 · Trigonometric functions and equations · spec MM4.6

Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?

Answer: F

$12(1 - \sin^2 x) + 6\sin x - 12 = 0 \Rightarrow 12\sin^2 x - 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0$. $\sin x = 0$ gives 180° (the endpoints 0° and 360° are excluded); $\sin x = \frac{1}{2}$ gives 30° and 150° .

Full method - how and why

Another quadratic in $\sin x$, but this one has no constant term, so factorising gives $\sin x = 0$ as well - and the interval is open.

- 1 Replace $\cos^2 x$: $12(1 - \sin^2 x) + 6\sin x - 10 = 2$.

why: Convert to a single trig function.

- 2 Tidy: $12 - 12\sin^2 x + 6\sin x - 12 = 0 \Rightarrow -12\sin^2 x + 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0$.

why: The constants cancel completely, leaving a **factorisable** expression with no constant term - do not divide by $\sin x$, or you lose solutions.

- 3 $\sin x = 0$ gives $x = 180^\circ$ within the open interval (0° and 360° are excluded).

why: The interval is $0^\circ < x < 360^\circ$, strict at both ends - so the endpoints are out but 180° is in.

- 4 $\sin x = \frac{1}{2}$ gives $x = 30^\circ$ and 150° .

why: First and second quadrants.

- 5 Complete set: $30^\circ, 150^\circ, 180^\circ$.

why: Three solutions - the 180° is what distinguishes the correct option.

Where it goes wrong

- Dividing by $\sin x$ and losing 180° , giving just $30^\circ, 150^\circ$ - an offered option.
- Including 0° and 360° , which the strict inequalities exclude.

Takeaway Never divide by a trig function: factorise, so the zero solutions survive; then read the interval's endpoints carefully.

Question 5 TRG-15 · Trigonometric functions and equations · spec MM4.4

The graph of $y = \sin x$ (with x in radians) crosses the line $y = x$ at how many points?

Answer: D

$|\sin x| < |x|$ for all $x \neq 0$, so the only intersection is at the origin.

Full method - how and why

A graphical comparison. The line $y = x$ grows without bound while $\sin x$ is trapped between -1 and 1 .

- 1 Note $x = 0$ is a solution: $\sin 0 = 0$.

why: Both graphs pass through the origin.

- 2 For $x > 0$, $\sin x < x$ always (the sine curve leaves the origin with gradient 1 and then bends away below the line).

why: The gradient of $\sin x$ is at most 1, and it is strictly less than 1 after the origin, so the line pulls ahead immediately.

- 3 For $x < 0$, by symmetry (both functions are odd), $\sin x > x$.

why: Odd symmetry means the picture is rotated, not reflected - so again no crossing.

- 4 Hence exactly one intersection, at the origin.

why: The answer is 1.

Where it goes wrong

- Answering 'infinitely many' by confusing $y = x$ with a horizontal line like $y = 0.5$.
- Thinking there must be crossings because the sine oscillates - it oscillates, but only within $[-1, 1]$.

Takeaway $|\sin x| \leq 1$ while $|x|$ grows: only the origin can satisfy $\sin x = x$.

Question 6 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$: $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.

Full method - how and why

Three identity claims. Two are standard, one is a classic false 'linearity' claim.

1 I: $\sin^2 x + \cos^2 x = 1$ is the Pythagorean identity, true for every x . **True.**

why: It follows from Pythagoras applied to the unit circle.

2 II: test $x = \frac{\pi}{2}$: $\sin \pi = 0$ but $2\sin \frac{\pi}{2} = 2$. **False.**

why: Trig functions are not linear: $\sin(2x) \neq 2\sin x$ (the correct identity, off-specification here, is $2\sin x \cos x$).

3 III: shifting $\sin$ by half a period reflects it: $\sin(x + \pi) = -\sin x$. **True.**

why: Check at $x = 0$: $\sin \pi = 0 = -\sin 0$ ✓; at $x = \frac{\pi}{2}$: $\sin \frac{3\pi}{2} = -1 = -\sin \frac{\pi}{2}$ ✓.

4 I and III only.

why: Match to the options.

Where it goes wrong

- Accepting II because it 'looks like' a factor coming out.
- Doubting III - test it at two values to be sure.

Takeaway Functions are almost never linear: $f(2x) \neq 2f(x)$ in general. Test suspicious identities at $x = \frac{\pi}{2}$.

Question 7 TRG-10 · Trigonometric functions and equations · spec MM4.6

How many solutions does the equation $\sin x = \frac{x}{10}$ have?

Answer: A

Solutions need $|x| \leq 10$. Both sides are odd functions, so count $x > 0$ and double, then add $x = 0$. On $(0, \pi)$ the curve starts above the line (since $\sin x \approx x$) and ends below it: one crossing. On $(\pi, 2\pi)$, $\sin x < 0$: none. On $(2\pi, 3\pi)$ the line is between 0.63 and 0.94 while $\sin x$ rises to 1: two crossings ($3\pi \approx 9.42 < 10$). Beyond $4\pi > 10$ the line exceeds 1. So 3 positive, 3 negative and 0: seven solutions.

Full method - how and why

A transcendental equation - it cannot be solved algebraically. Count intersections of $y = \sin x$ with the straight line $y = \frac{x}{10}$ by sketching.

- 1 Bound the solutions: $|\sin x| \leq 1$, so $|\frac{x}{10}| \leq 1$, i.e. $-10 \leq x \leq 10$.

why: Outside that range the line is beyond the reach of the sine curve, so no intersections are possible.

- 2 Note both sides are odd functions, so intersections come in $\pm$ pairs; count for $x > 0$ and double, then add $x = 0$ (which is a solution).

why: $\sin(-x) = -\sin(x)$... more simply, if x_0 is a solution so is $-x_0$, because both sides change sign together.

- 3 For $0 < x \leq 10$: on $(0, \pi)$ the sine hump starts steeper than the line and ends below it, giving **one** crossing. On $(\pi, 2\pi)$ the sine is negative while the line is positive: **none**.

why: Near 0, $\sin x$ rises much faster than $\frac{x}{10}$; by $x = \pi$ it has returned to 0, below the line's value 0.31.

- 4 On $(2\pi, 3\pi)$ (i.e. up to 9.42) the line runs from 0.63 to 0.94 while the sine hump rises to 1 and falls back: **two** crossings. Beyond 3π the next hump starts past $x = 4\pi > 10$.

why: A hump that rises above the line and comes back down crosses it twice.

- 5 So 3 positive solutions, 3 negative and $x = 0$: **7** in total.

why: Doubling the positive count and adding the origin.

Where it goes wrong

- Forgetting $x = 0$ and answering 6.
- Missing that the hump near $x = 8$ crosses the line **twice**, and answering 5.

Takeaway For $\sin x = kx$, sketch both graphs, bound the region by $|kx| \leq 1$, and use oddness to halve the counting.